



BRAC UNIVERSITY
Department of Computer Science & Engineering
MIDTERM EXAMINATION, Fall 2025
CSE 350: Digital Electronics and Pulse Techniques



Total Marks: 50 (20 + 15 + 15)

Time: **90 Minutes**

- Total no of questions are **four (4)**. Answer **them all**. **Question 4** is a bonus question.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet.

Name:	ID:	Section:
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■ **Question 1** [CO3] [20 marks]

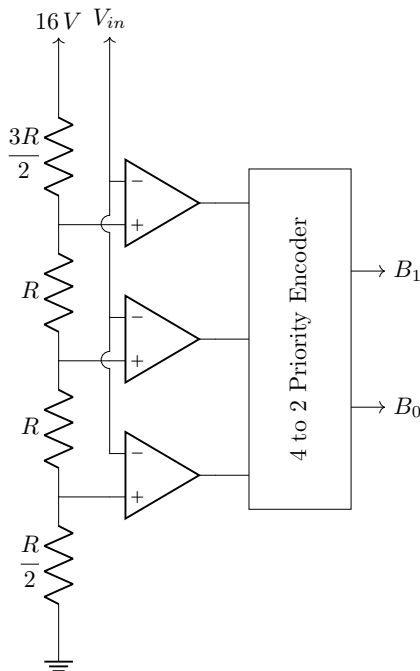


Figure 1: 2-bit Flash ADC

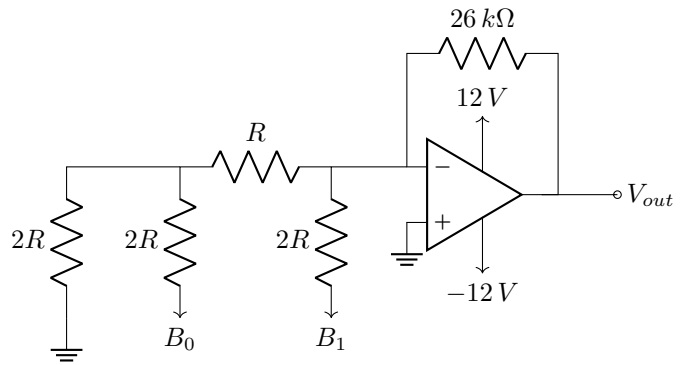


Figure 2: 2-bit R-2R Ladder DAC

$$V_{out} = -\frac{R_f}{2R} \left(V_{B_1} + \frac{V_{B_0}}{2} \right)$$

For both the circuits in the above figures, voltage of B_1 & B_0 is 5V when logic is **HIGH**, and 0V, when logic is **LOW**. The output of the ADC is passed to the DAC as input.

- Make a table listing the **quantization intervals** of the ADC shown in figure 1, along with the respective **digital output**. [4]
- If the maximum digital input to the DAC, produces a V_{out} of **-13.0 V**, find the value of the resistor 'R' in figure 2. Then calculate the **step size** of the DAC, and list all the output voltages for the **remaining inputs**. [4]
- Using your results from (a) & (b), find the digital output of the ADC for $V_{in} = 11V$ and use it to find the V_{out} of the DAC. [3]

Now consider a 2-bit **Dual Slope ADC** with a similar reference voltage as in figure 1, was used instead of the Flash ADC. It has a clock frequency of 4 MHz .

- (d) Find the input voltage V_{in} for which the conversion time was found to be $1.35\text{ }\mu\text{s}$. Also find the **maximum conversion time** of the ADC. [4]
- (e) If for the Dual Slope ADC $V_{in} = 11\text{ V}$, find its digital output ' B_1B_0 '. Then find V_{out} if this was passed to the DAC in figure 2. Find the **%error** between the values of V_{out} for '1(c)' & '1(e)'. [5]

■ **Question 2** [CO3] [15 marks]

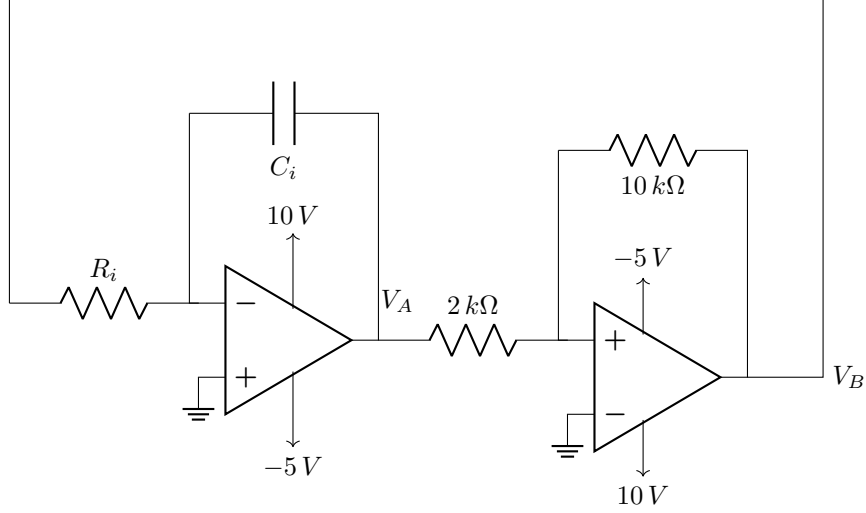


Figure 3: A Triangular Wave Generator

- (a) For the triangular wave generated by the circuit in figure 3, prove that $f = \frac{10}{9R_iC_i}$. [6]
- (b) You have to design a **square wave generator** that can generate a square wave having duty cycle similar to the square wave at V_B . Find V_H , V_L and draw the circuit given that, $R_1 = 10\text{ k}\Omega$, $R_2 = 20\text{ k}\Omega$, $R_x = 10\text{ k}\Omega$, $C_x = 50\text{ nF}$ and $f = 1\text{ kHz}$. Here, $\tau = R_xC_x$ and R_2 is the feedback resistance. [Hint: You may need to assume one of the bias voltages] [9]

■ **Question 3** [CO3] [15 marks]

The input to & output from a certain Schmitt Trigger is shown in figure 4.

- (a) From figure 4, find the **bias voltages** & the **threshold voltages** of the Schmitt Trigger. Then draw its **VTC** (Voltage Transfer Characteristics) on the plot given in figure 5. [5]
- (b) Find the **shift voltage**. Design the Schmitt Trigger by specifying the values of each component and drawing the full circuit. [5]
- (c) If the connections of **input voltage** & **reference voltage** in the circuit of 'b' are swapped, find the new **thresholds**, **shift voltage** & the **hysteresis width**. What type of Schmitt Trigger is this modified circuit? [5]

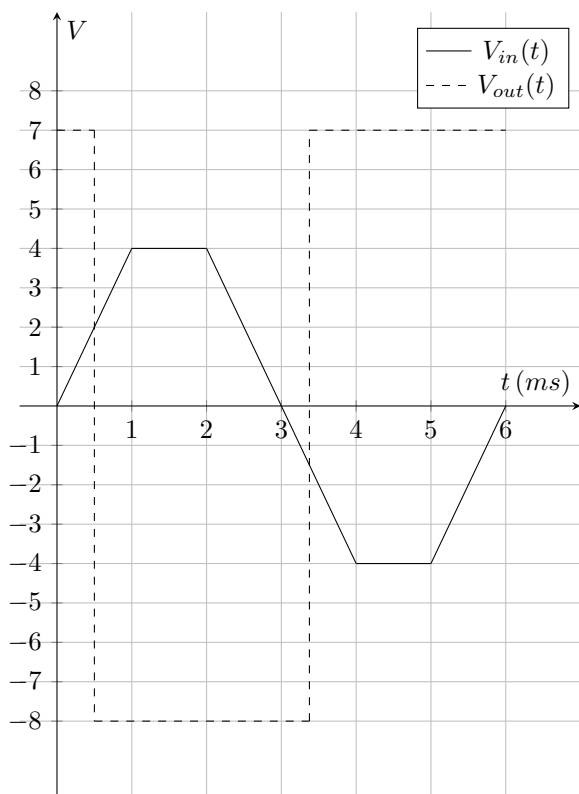


Figure 4: $V_{in}(t)$ & $V_{out}(t)$

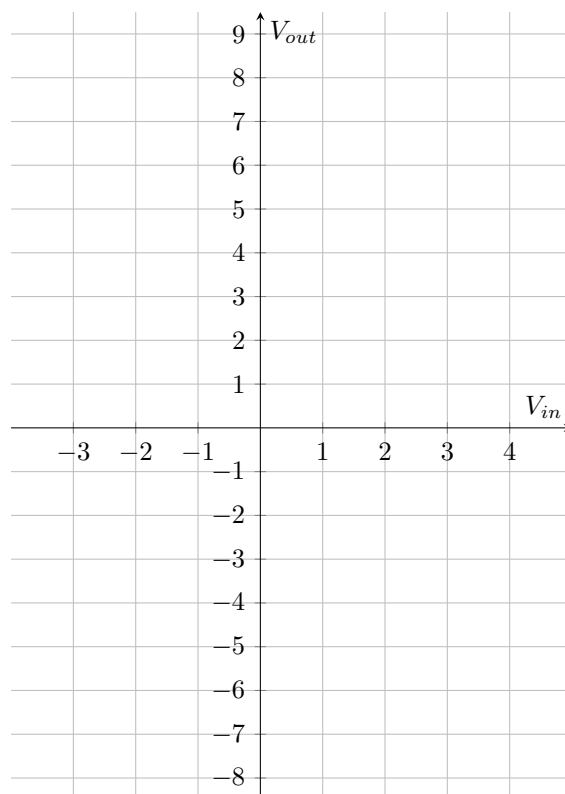


Figure 5: VTC of the Schmitt Trigger

■ **Question 4 [Bonus]** [CO3] [5 marks]

Answer the following questions in short (1-2 lines are enough).

- Can we design a Schmitt Trigger with a hysteresis width of **0 V**? [1]
- Why is the integration time (t_1) always same for a Dual Slope ADC even for different input voltages? Furthermore, why does it take different times in the de-integration phase despite the reference voltage being same for all inputs? [2]
- From where can we get the following waves? [2]

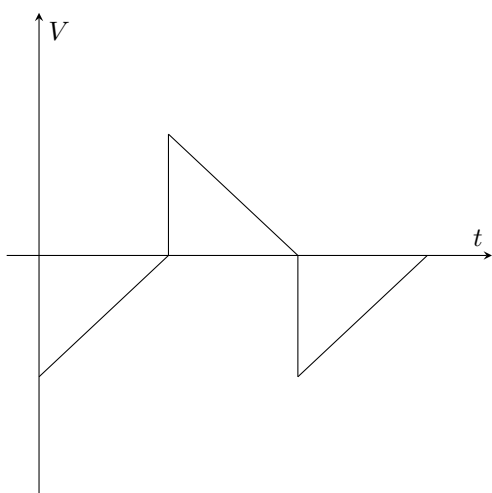


Figure 6: Wave A

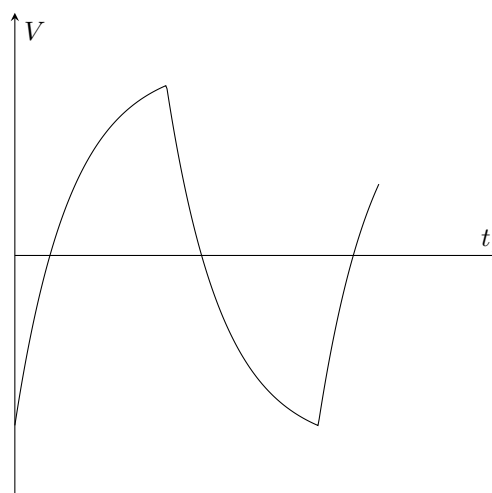


Figure 7: Wave B